IGCSE 0607 Practice Questions — E8.1 — Combined Questions & Markscheme

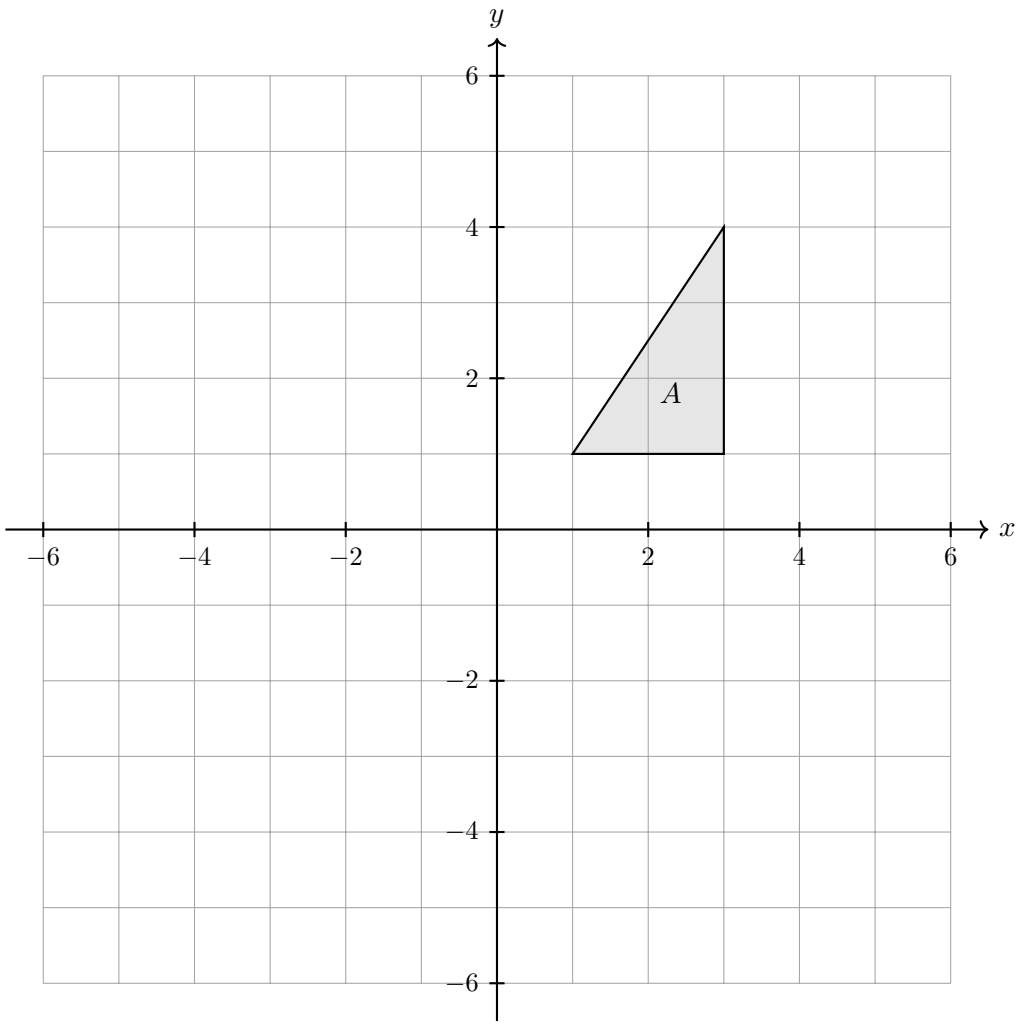
Transformations

Total Questions: 24 — **Total Marks:** 158 Marks

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- **E8.1** Transformations
  → **Paper 2 (Non-Calculator)** *(13 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

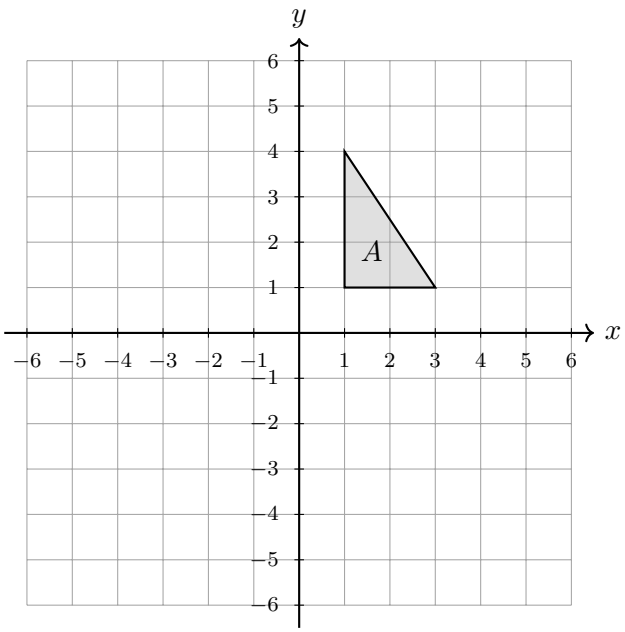
1 The grid shows triangle A.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $x = -1$.
Label the image C. **[2]**

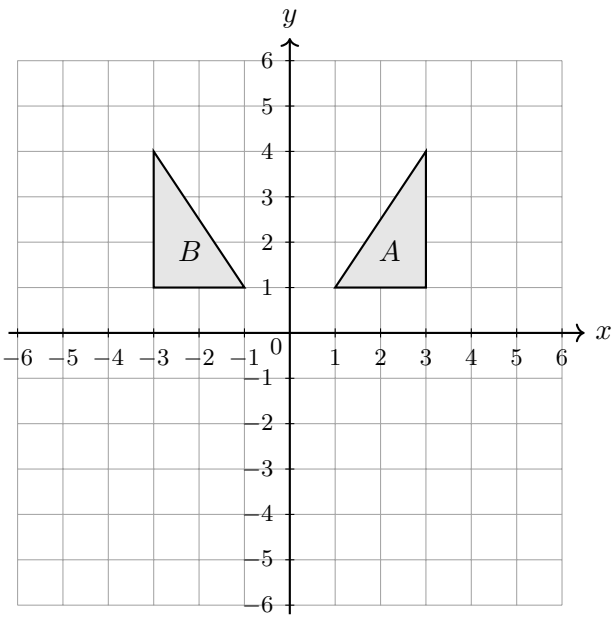
2 The diagram shows a triangle, Shape A, drawn on a coordinate grid.



(a) On the grid, draw the image of Shape A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$.
Label this image B. **[2]**

(b) On the grid, draw the image of Shape A after a reflection in the line $y = -1$.
Label this image C. **[2]**

3 The diagram shows two triangles, A and B, plotted on a coordinate grid.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$. **[2]**

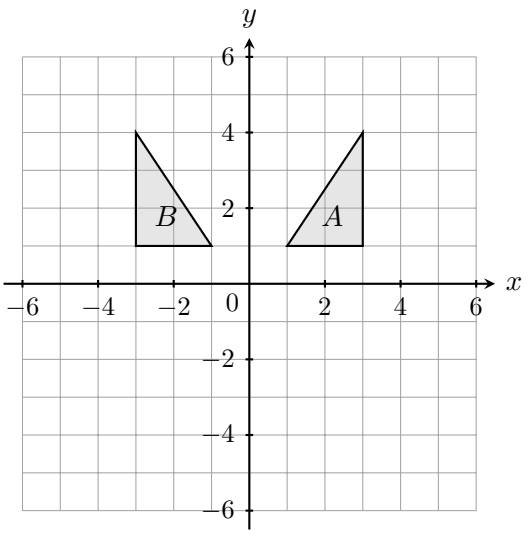
(b) Describe fully the single transformation that maps triangle A onto triangle B.

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..... **[2]**

4 The grid shows triangle A and triangle B.



(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -2 \\ -5 \end{pmatrix}$. **[2]**

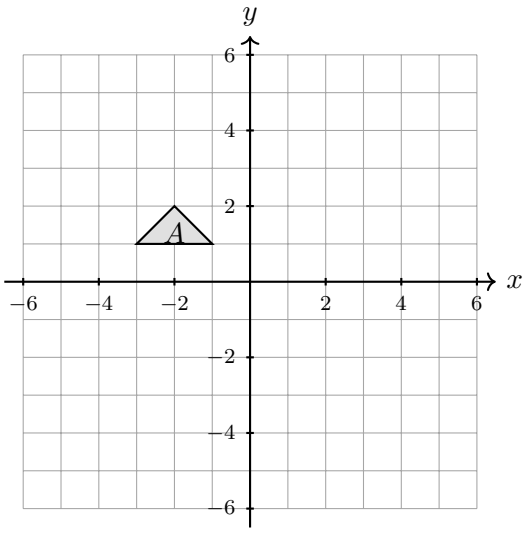
(b) Describe fully the single transformation that maps triangle A onto triangle B.

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.....

[2]

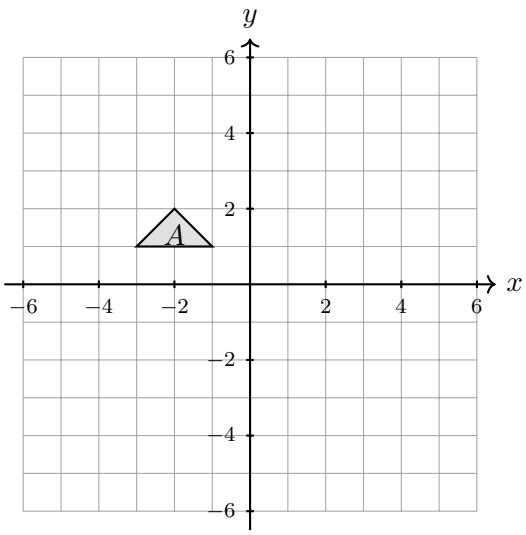
5 A triangle A is drawn on a grid.

(a) On the grid above, draw the image of triangle A after a translation by the vector $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.



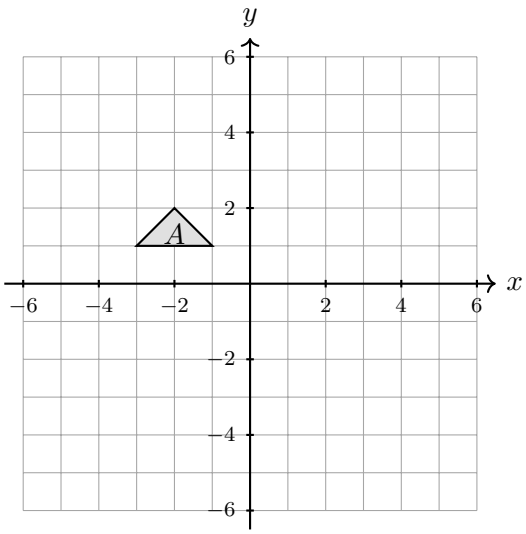
[2]

(b) On the grid above, draw the image of triangle A after a reflection in the line $y = x$.



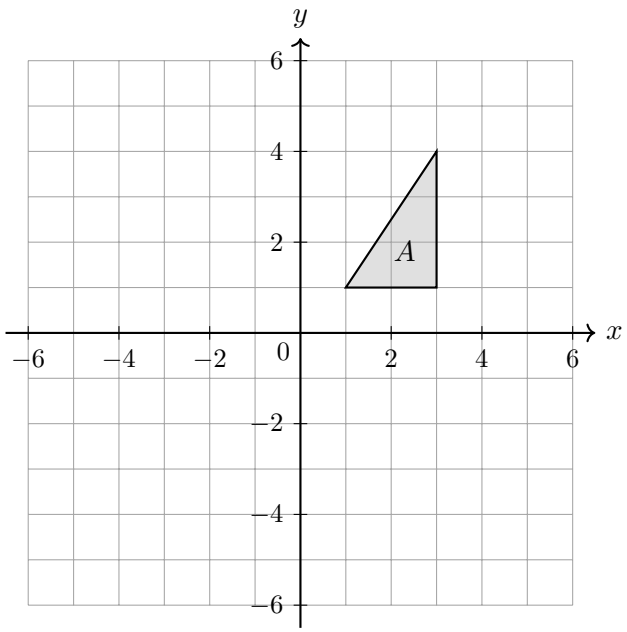
[2]

(c) On the grid above, draw the image of triangle A after an enlargement with scale factor 2 and centre $(-1, 0)$.



[2]

6 The diagram shows triangle A on a coordinate grid.

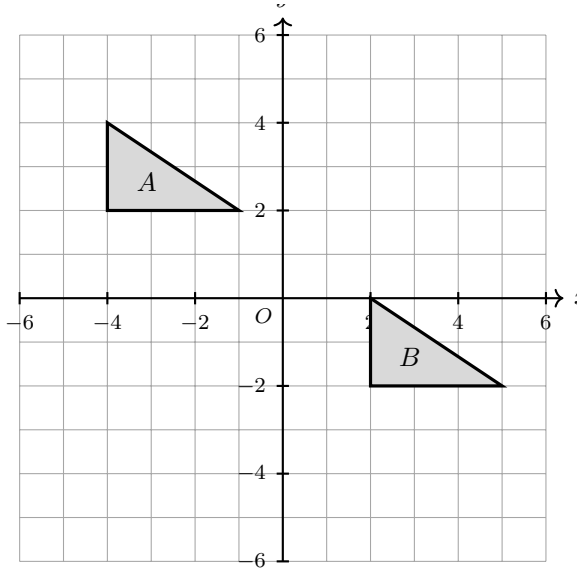


(a) On the grid, translate shape A by the vector $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$.
Label the image B. **[2]**

(b) On the grid, reflect shape A in the x-axis.
Label the image C. **[2]**

(c) On the grid, rotate shape A by 180° about the origin $(0, 0)$.
Label the image D. **[2]**

7 The diagram shows triangles A and B on a coordinate grid.



(a) Describe fully the single transformation that maps triangle A onto triangle B.

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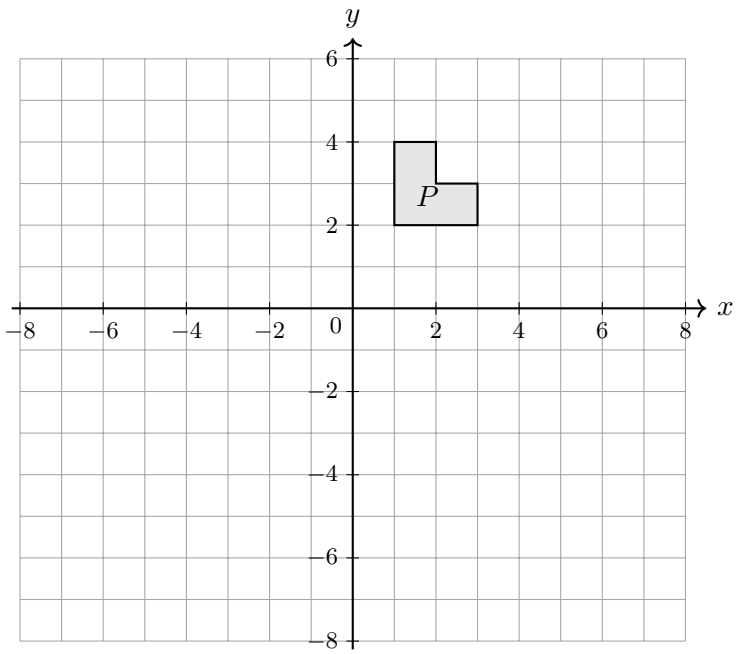
.....

..... **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $y = 1$.
Label the image C. **[2]**

(c) On the grid, draw the image of triangle A after a rotation of 90° clockwise about the origin $(0, 0)$.
Label the image D. **[3]**

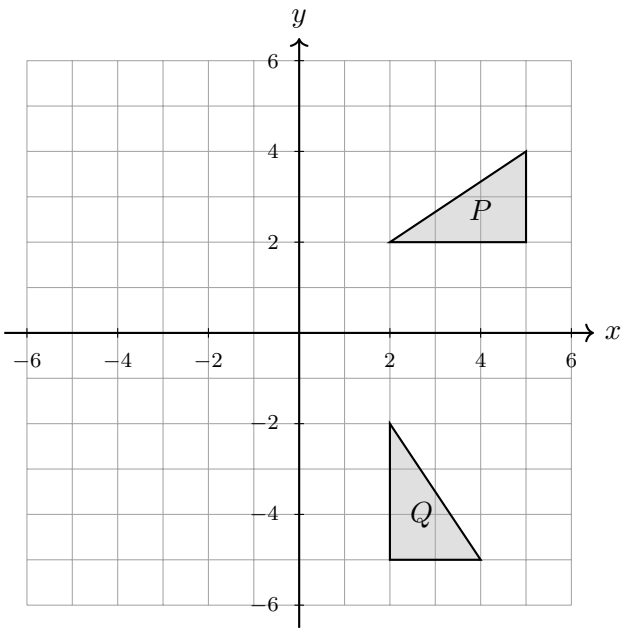
8 The diagram shows a shape P plotted on a coordinate grid.



(a) On the grid, draw the image of shape P after an enlargement, scale factor -2, centre $(0, 0)$. **[2]**

(b) On the grid, draw the image of shape P after a rotation of $90°$ clockwise about the point $(1, -1)$. **[2]**

9 The diagram shows two triangles, Shape P and Shape Q, drawn on a grid.



(a) Describe fully the single transformation that maps Shape P onto Shape Q.

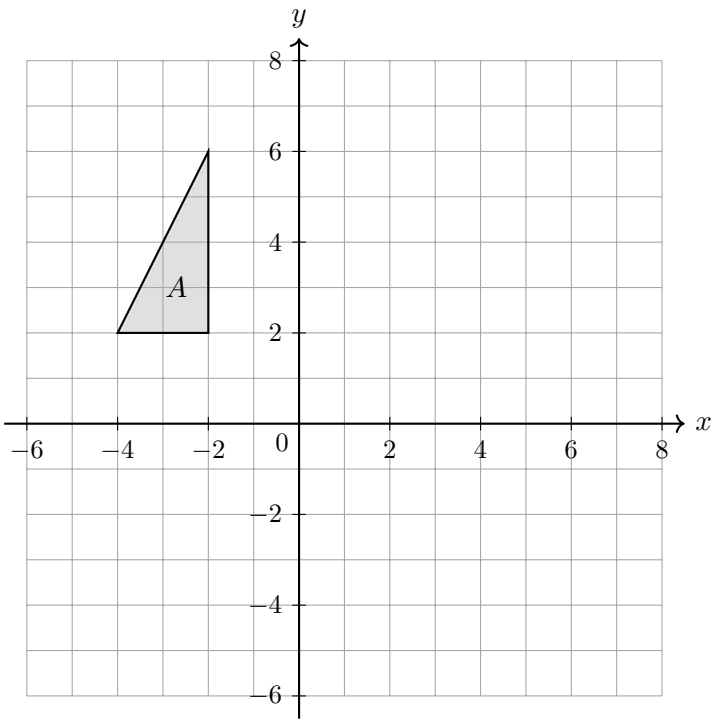
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..... **[3]**

(b) Shape P is enlarged by a scale factor of $\frac{1}{2}$ with centre of enlargement $(-2, 2)$.
On the grid, draw the image of Shape P and label it R. **[2]**

10 The diagram shows shape A on a coordinate grid.



(a) Shape A is reflected in the line $y = x$ to give shape B.
On the grid, draw and label shape B. **[2]**

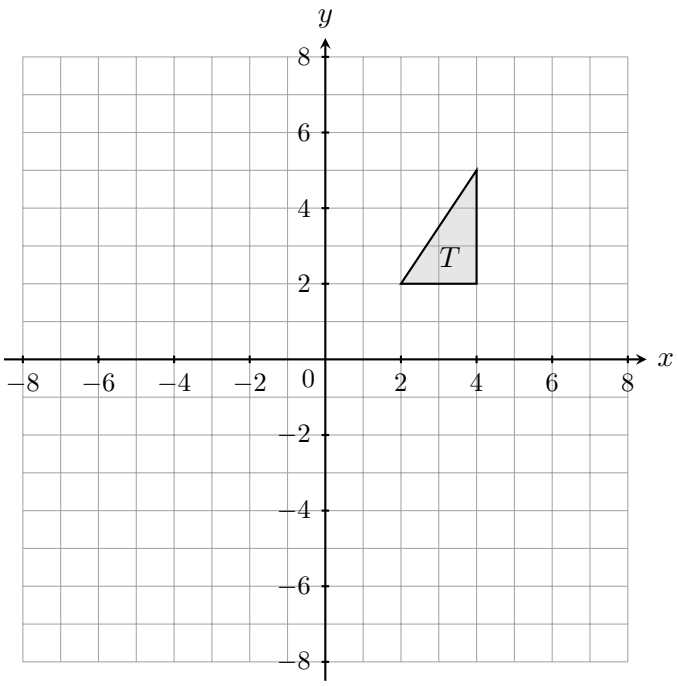
(b) Shape B is rotated by 90° clockwise about the point $(2, 2)$ to give shape C.
On the grid, draw and label shape C. **[2]**

(c) Describe fully the single transformation that maps shape A onto shape C.

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..... **[2]**

11 The grid shows triangle T.



(a) On the grid, draw the image of triangle T after an enlargement with scale factor $-\frac{1}{2}$ and centre $(0, 2)$. **[2]**

(b) Triangle T is reflected in the line $x = 1$ to give triangle U.
Triangle U is then reflected in the line $y = 2$ to give triangle V.
Describe fully the single transformation that maps triangle T onto triangle V.

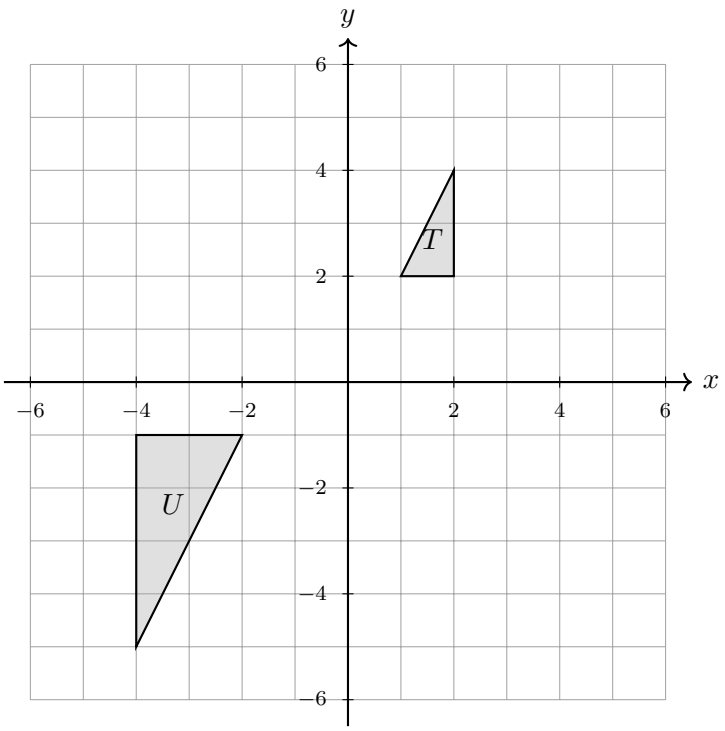
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[3]

(c) Triangle T is enlarged by a scale factor of k to give triangle W.
The area of triangle W is 48 square units.
Find the value of k, given that $k > 0$.

$k =$ ..... **[2]**

12 The diagram shows two triangles, Shape T and Shape U.



(a) Describe fully the single transformation that maps Shape T onto Shape U.

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..... **[3]**

(b) Describe fully the single transformation that maps Shape U onto Shape T.

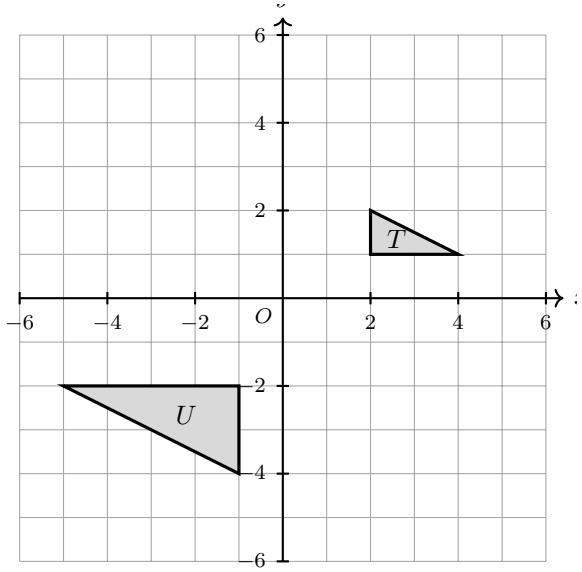
.....

..... **[2]**

(c) Shape T is translated by the vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$ to give Shape V.
Shape V is then reflected in the line $y = x$ to give Shape W.
Find the coordinates of the three vertices of Shape W.

..... **[3]**

13 The diagram shows triangles T and U on a coordinate grid.



(a) Describe fully the single transformation that maps triangle T onto triangle U.

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..... **[3]**

(b) Triangle T is reflected in the line $x = 1$ to give triangle V.
Triangle V is then reflected in the line $y = 2$ to give triangle W.

(i) On the grid, draw and label triangle W. **[3]**

(ii) Describe fully the single transformation that maps triangle T directly onto triangle W.

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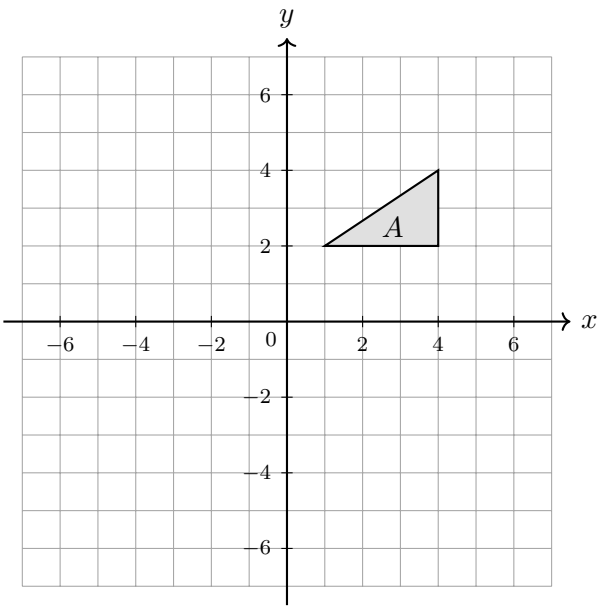
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[3]

(c) Point $P(a, b)$ is mapped onto the point $P'(5, -3)$ by a reflection in the line $y = x$ followed by a translation by the vector $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$.
Find the coordinates of P.

$P = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[3]**

14 The diagram shows triangle A on a coordinate grid.

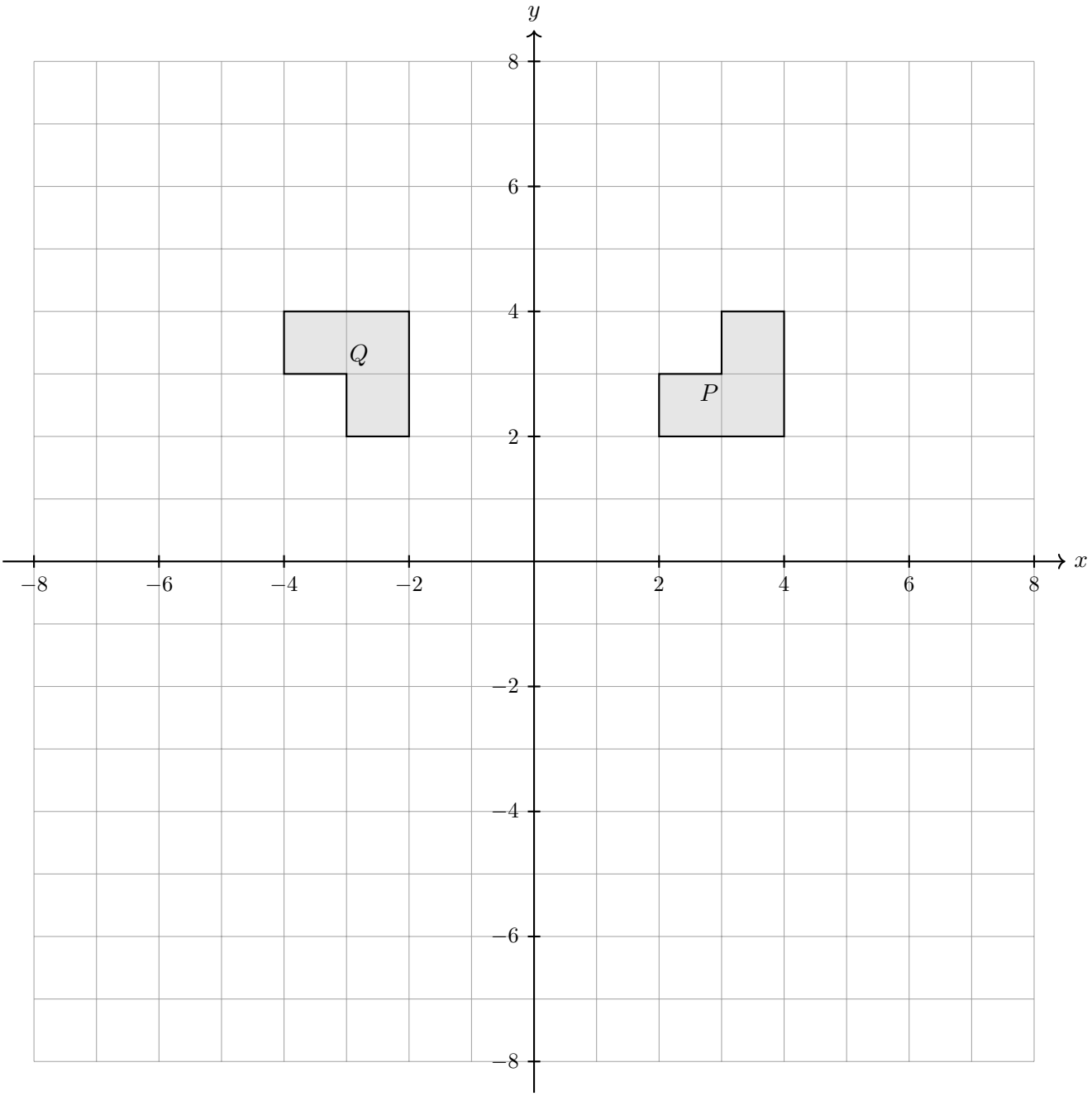


(a) On the grid, draw the image of triangle A after a translation by the vector $\begin{pmatrix} -6 \\ -3 \end{pmatrix}$. **[2]**

(b) On the grid, draw the image of triangle A after a reflection in the line $y = 1$. **[2]**

(c) On the grid, draw the image of triangle A after a rotation of $180°$ about the origin $(0, 0)$. **[2]**

15 The grid shows shapes P and Q.



(a) Describe fully the single transformation that maps shape P onto shape Q.

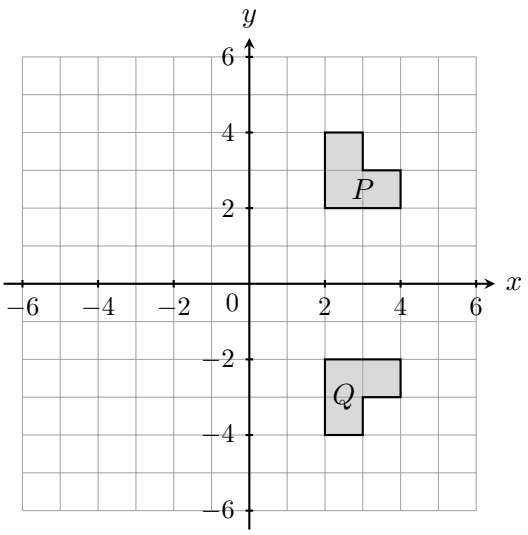
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..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$, centre $(6, 6)$.
Label the image R. **[2]**

16 The grid shows shapes P and Q.



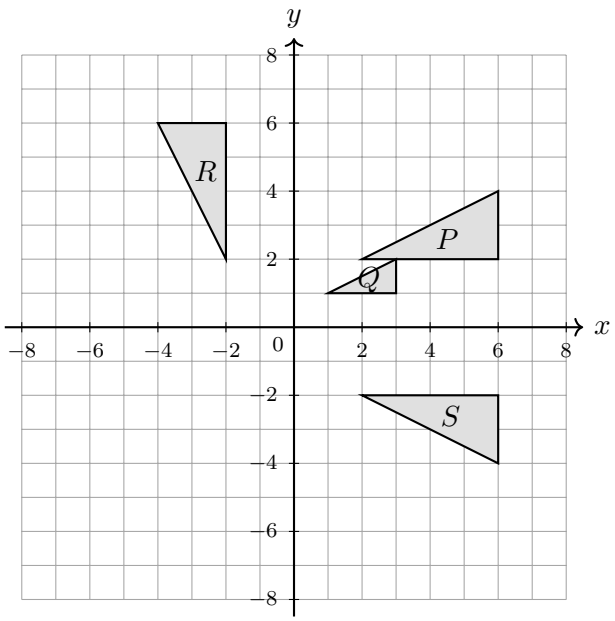
(a) On the grid, draw the image of shape P after an enlargement with scale factor 0.5 and centre $(0, 0)$. **[2]**

(b) Describe fully the single transformation that maps shape P onto shape Q.

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[3]

17 The diagram shows triangles P, Q, R, and S on a coordinate grid.



(a) Describe fully the single transformation that maps triangle P onto triangle Q.

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..... **[3]**

(b) Describe fully the single transformation that maps triangle P onto triangle R.

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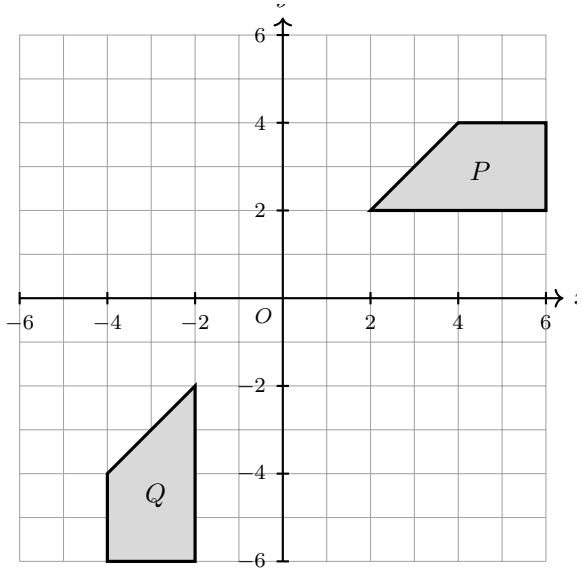
..... **[3]**

(c) Describe fully the single transformation that maps triangle P onto triangle S.

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..... **[2]**

18 The diagram shows shapes P and Q on a coordinate grid.



(a) Describe fully the single transformation that maps shape P onto shape Q.

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..... **[2]**

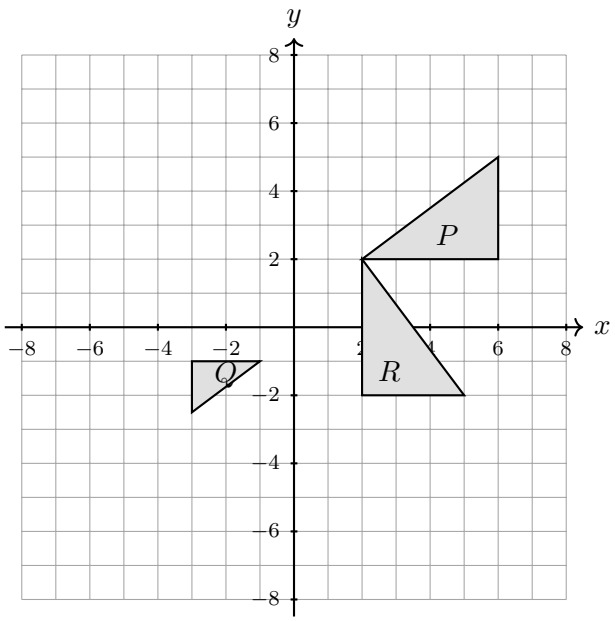
(b) On the grid, draw the image of shape P after an enlargement with scale factor $\frac{1}{2}$ and centre $(-2, -2)$.
Label the image R. **[3]**

(c) On the grid, draw the image of shape P after a rotation of 90° anticlockwise about the point $(1, 1)$.
Label the image S. **[3]**

(d) Find the ratio of the area of shape R to the area of shape P, giving your answer in its simplest form.

..... **[1]**

19 The diagram shows triangles P, Q and R.



(a) Describe fully the single transformation that maps triangle P onto triangle Q.

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..... **[3]**

(b) Describe fully the single transformation that maps triangle P onto triangle R.

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..... **[3]**

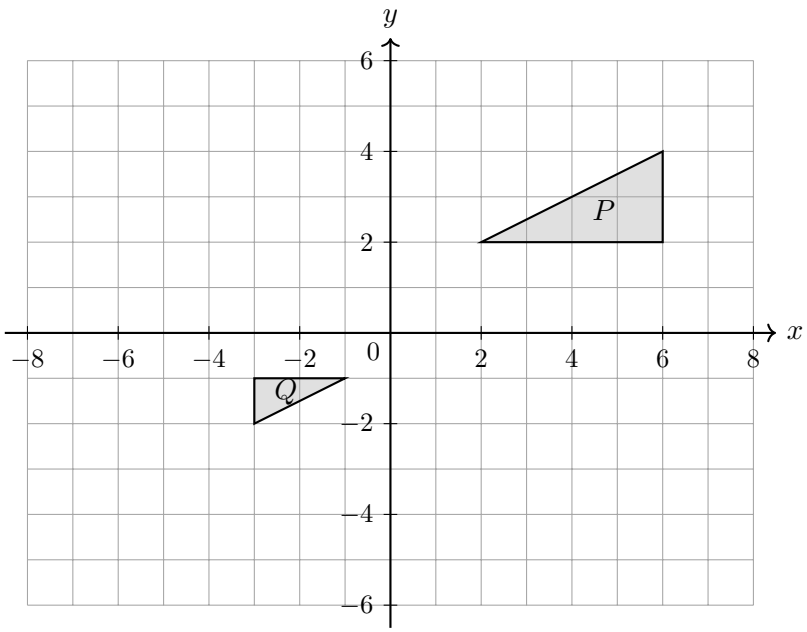
(c) Triangle Q is reflected in the origin $(0, 0)$ to give triangle S.
Describe fully the single transformation that maps triangle P onto triangle S.

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..... **[3]**

20 The diagram shows triangles P and Q on a coordinate grid.



(a) Describe fully the single transformation that maps shape P onto shape Q.

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..... **[3]**

(b) On the grid, enlarge shape P by scale factor $\frac{1}{2}$ with centre $(0, 4)$.
Label the image R. **[3]**

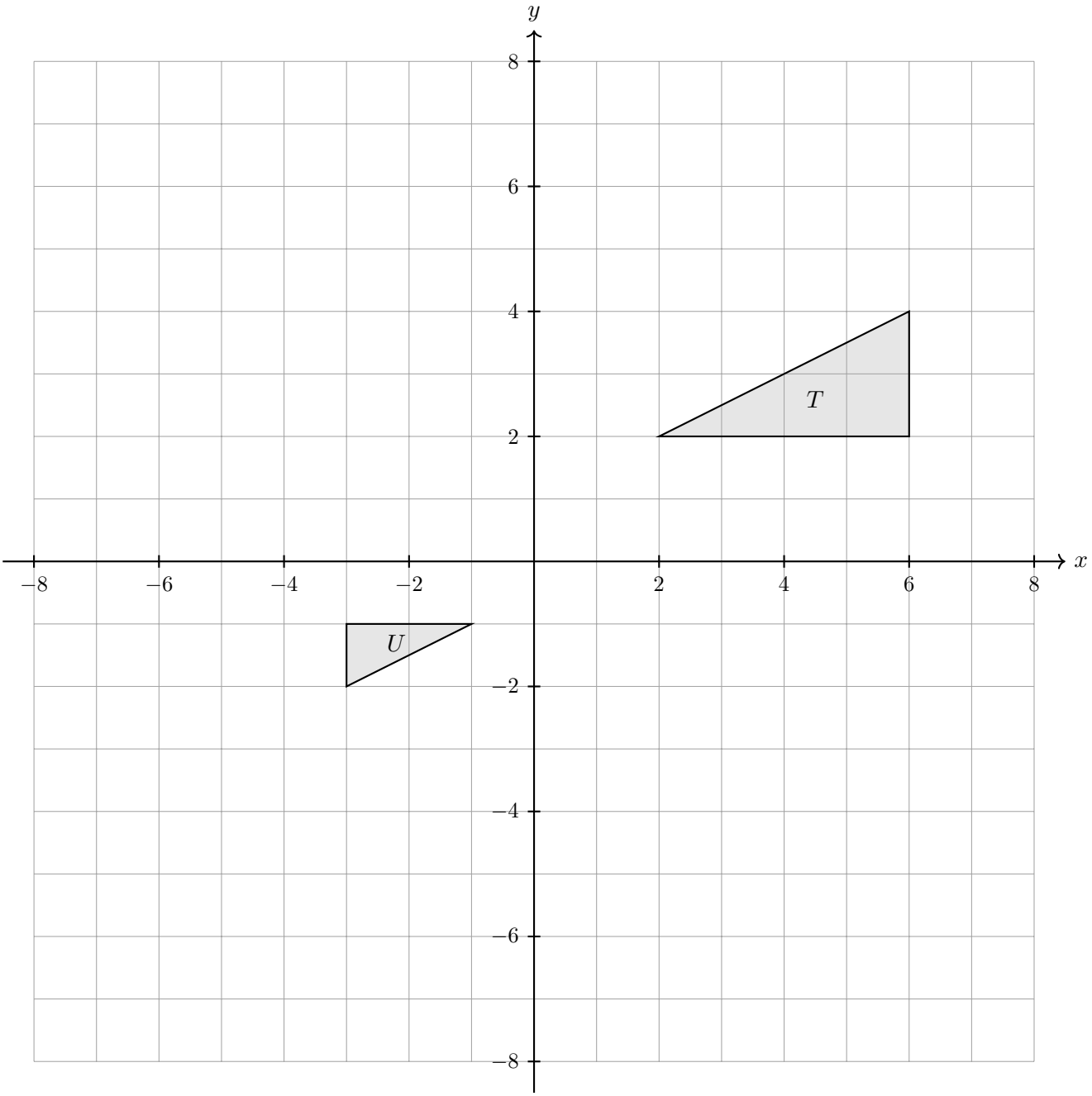
(c) Describe fully the single transformation that maps shape R onto shape Q.

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..... **[3]**

21 The grid shows triangles T and U.



(a) Describe fully the single transformation that maps triangle T onto triangle U.

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..... **[3]**

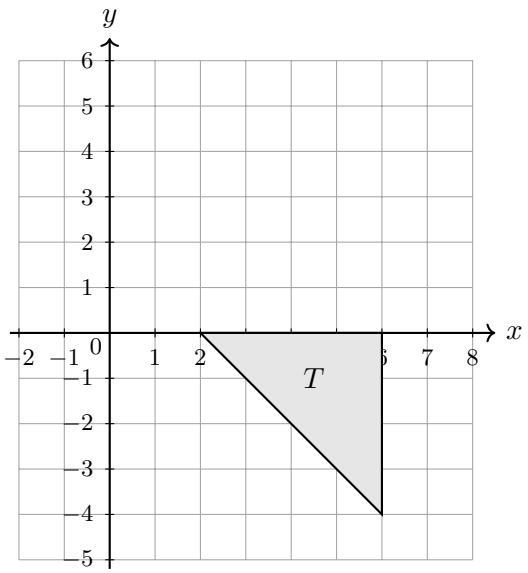
(b) Triangle T is reflected in the line $y = x$ to give triangle V.
Triangle V is then rotated 90° clockwise about the origin to give triangle W.
Describe fully the single transformation that maps triangle T onto triangle W.

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..... **[2]**

22 The diagram shows triangle T plotted on a coordinate grid.



(a) Triangle T is mapped onto triangle U by an enlargement, scale factor $\frac{1}{2}$, centre $(0, 4)$.
On the grid, draw triangle U. **[2]**

(b) Triangle U is mapped onto triangle V by a reflection in the line $y = x$.
On the grid, draw triangle V. **[2]**

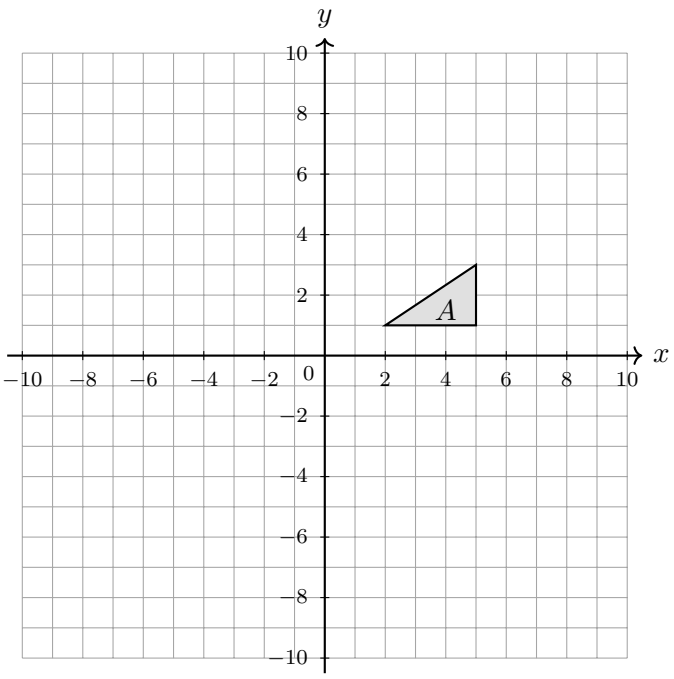
(c) Describe fully the single transformation that maps triangle U back onto triangle T.

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..... **[3]**

23 The diagram shows triangle A on a coordinate grid.



(a) On the grid, draw the image of triangle A after an enlargement by scale factor -2, centre $(1, 0)$. Label this image B. **[3]**

(b) Triangle A is mapped onto triangle C by a reflection in the line $y = -x$.
Write down the coordinates of the vertices of triangle C.

..... **[2]**

(c) Triangle A is first translated by the vector $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$.

The resulting image is then rotated 90° clockwise about the point $(5, 5)$ to form triangle D.
Describe fully the single transformation that maps triangle A onto triangle D.

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..... **[3]**

24 A triangle T has vertices at (a, b), (c, d) and (e, f).
The translation by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ maps triangle T onto triangle U with vertices at $(5, 2)$, $(8, 2)$ and $(7, 6)$.

(a) Find the coordinates of the three vertices of triangle T.

..... **[2]**

(b) Triangle U is mapped onto triangle V by a reflection in the line $y = x$.
Write down the coordinates of the three vertices of triangle V.

..... **[2]**

(c) Triangle V is mapped onto triangle W by a rotation of $90°$ clockwise about the point $(2, 5)$.
Find the coordinates of the three vertices of triangle W.

..... **[3]**

(d) Describe fully the single transformation that maps triangle T onto triangle X, where the vertices of X are $(-4, -8)$, $(-10, -8)$ and $(-8, -16)$.

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.....

..... **[3]**

1 **(a)** Correct triangle B with vertices at $(-3,3),(-1,3),(-1,6)$ **[B2]**

Translation by vector $\begin{pmatrix}-4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$ **[B1]**

(b) Correct triangle C with vertices at $(-3,1),(-5,1),(-5,4)$ **[B2]**

Reflection in $y=-1$ or correct orientation but wrong position **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Correct translation of Shape A **[B2]**

If incorrect, translation of $\begin{pmatrix}-4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$ where $k\neq 0$ **[B1]**

(b) Correct reflection of Shape A to form Shape C with vertices at $(1,-3)$, $(3,-3)$, and $(1,-6)$ **[B2]**

If incorrect, reflection in $x=-1$ or $y=1$ **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Correct triangle with vertices at $(-3,3),(-1,3)$ and $(-1,6)$ **[B2]**

Translation by correct vector but incorrect shape/orientation OR correct size, shape and orientation but wrong translation **[B1]**

(b) Reflection **[B1]**

in the y-axis OR $x=0$ **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Vertices of image plotted correctly at $(-1,-4)$, $(1,-4)$ and $(1,-1)$ **[B2]**

If not B2, award B1 for correct size and orientation, or 2 vertices correct **[B1]**

(b) Reflection **[B1]**

in the y-axis (or $x=0$) **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Vertices of the image at $(3,-2),(1,-2),(2,-1)$ **[B2]**

Correct orientation but wrong position, or translation by $\begin{pmatrix}4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\-3\end{pmatrix}$ **[B1]**

(b) Vertices of the image at $(1,-1),(1,-3),(2,-2)$ **[B2]**

Reflection in $y=-x$ or reflection in x-axis or y-axis **[B1]**

(c) Vertices of the image at $(-1,2),(-5,2),(-3,4)$ **[B2]**

Correct size and orientation but wrong position, OR enlargement with SF 2 from a different centre **[B1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Image B drawn correctly at $(-4,3),(-2,3),(-2,6)$ **[B2]**

If incorrect, 1 mark for translation by $\begin{pmatrix}-5\\k\end{pmatrix}$ or $\begin{pmatrix}k\\2\end{pmatrix}$ **[M1]**

(b) Image C drawn correctly at $(1,-1),(3,-1),(3,-4)$ **[B2]**

If incorrect, 1 mark for reflection in y-axis or in a line $y=k$ **[M1]**

(c) Image D drawn correctly at $(-1,-1),(-3,-1),(-3,-4)$ **[B2]**

If incorrect, 1 mark for rotation of 180° about the wrong centre, or 90° about $(0,0)$ **[M1]**

[Back to Question] *[Total Marks: 6]*

7 **(a)** Translation **[B1]**

By vector $\begin{pmatrix}6\\-4\end{pmatrix}$ **[B1]**

(b) Correct reflection drawn with vertices at $(-4,-2)$, $(-4,0)$, and $(-1,0)$ **[B2]**

Award [B1] if reflected correctly in any horizontal line $y=k$ where $ke1$

(c) Correct rotation drawn with vertices at $(4,4)$, $(2,4)$, and $(2,1)$ **[B3]**

Award [B2] for correct orientation and size in wrong position, or 90° anticlockwise rotation about $(0,0)$

Award [B1] for any two vertices plotted correctly

[Back to Question] *[Total Marks: 7]*

8 **(a)** Correct shape with vertices at $(-2,-4),(-6,-4),(-6,-6),(-4,-6),(-4,-8)$ and $(-2,-8)$ **[B2]**

Enlargement, scale factor 2 centre $(0,0)$ OR correct scale factor and orientation but wrong centre **[B1]**

(b) Correct shape with vertices at $(4,-1),(4,-3),(5,-3),(5,-2),(6,-2)$ and $(6,-1)$ **[B2]**

Rotation of 90° clockwise about the wrong centre OR rotation of 90° anti-clockwise about $(1,-1)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** Rotation **[B1]**

90° clockwise OR 270° anti-clockwise **[B1]**

Centre $(0,0)$ **[B1]**

(b) Correct triangle drawn with vertices at $(0,2)$, $(1.5,2)$, and $(1.5,3)$ **[B2]**

If incorrect, correct size and orientation but wrong position OR correct centre of enlargement but scale factor 2 **[B1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** Image B drawn correctly at $(2,-4),(2,-2),(6,-2)$ **[B2]**

If incorrect, 1 mark for reflection in $y=-x$ or correct orientation but wrong position **[M1]**

(b) Image C drawn correctly at $(-4,2),(-2,2),(-2,-2)$ **[B2]**

If incorrect, 1 mark for rotation 90° anti-clockwise about $(2,2)$ or 90° clockwise about the wrong centre **[M1]**

(c) Reflection **[B1]**

in the line $y=2$ **[B1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Vertices of image plotted correctly at $(-1,2)$, $(-2,2)$ and $(-2,0.5)$ **[B2]**

If not B2, award B1 for enlargement SF $-\frac{1}{2}$ from wrong centre, or correct centre but SF $\frac{1}{2}$ **[B1]**

(b) Rotation **[B1]**

180° (direction not required) **[B1]**

centre $(1,2)$ **[B1]**

(c) Calculates area of $T=\frac{1}{2}\times 2\times 3=3$ **[M1]**

Sets up $3\times k^2=48 \implies k^2=16$ **[M1]**

$k=4$ **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0,1)$ **[B1]**

(b) Enlargement scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0,1)$ **[B1]**

(c) Translating Shape T gives at least two correct coordinates of V: $(-2,3)$, $(-1,3)$, $(-1,5)$ **[M1]**

Recognising reflection in $y=x$ swaps coordinates (x,y)

(y, x)$[M1]$

Final coordinates: $(3,-2)$, $(3,-1)$, $(5,-1)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

13 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(1,0)$ **[B1]**

(b) **(i)** Intermediate triangle V located with vertices at $(0,1)$, $(-2,1)$, and $(0,2)$ **[M1]**

Correct triangle W drawn with vertices at $(0,3)$, $(-2,3)$, and $(0,2)$ **[A2]**

Award [A1] for any two vertices of W plotted correctly

(ii) Rotation **[B1]**

Through $180°$ **[B1]**

About centre $(1,2)$ **[B1]**

(c) Reflection of (a,b) in $y=x$ gives (b,a) **[B1]**

$(b+2, a-4) = (5,-3)$ giving $b=3$ and $a=1$ **[M1]**

$(1,3)$ **[A1]**

[Back to Question] *[Total Marks: 12]*

14 **(a)** Correct triangle drawn with vertices at $(-5,-1)$, $(-2,-1)$, $(-2,1)$ **[B2]**

If not B2, B1 for correct size and orientation but wrong position OR two correct vertices **[B1]**

(b) Correct triangle drawn with vertices at $(1,0)$, $(4,0)$, $(4,-2)$ **[B2]**

If not B2, B1 for reflection in $x=1$ OR reflection in $y=k$ where $k \neq 1$ **[B1]**

(c) Correct triangle drawn with vertices at $(-1,-2)$, $(-4,-2)$, $(-4,-4)$ **[B2]**

If not B2, B1 for rotation of $90°$ about $(0,0)$ OR rotation of $180°$ about a different centre **[B1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** Rotation **[B1]**

$90°$ anticlockwise OR $270°$ clockwise **[B1]**

Centre $(0,0)$ **[B1]**

(b) Correct shape R with vertices at $(4,4), (5,4), (5,5), (4.5,5), (4.5,4.5), (4,4.5)$ **[B2]**

Any enlargement scale factor $\frac{1}{2}$ OR correct centre $(6,6)$ with incorrect scale factor **[B1]**

[Back to Question] *[Total Marks: 5]*

16 **(a)** Vertices of image plotted correctly at $(1,1)$, $(2,1)$, $(2,1.5)$, $(1.5,1.5)$, $(1.5,2)$ and $(1,2)$ **[B2]**

If not B2, award B1 for correct size and orientation, or correct enlargement from the wrong centre **[B1]**

(b) Rotation **[B1]**

$90°$ clockwise (or $-90°$) **[B1]**

centre $(0,0)$ (or origin) **[B1]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** Enlargement **[B1]**

Scale factor $\frac{1}{2}$ (or 0.5) **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Rotation **[B1]**

90° anticlockwise OR 270° clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(c) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Back to Question] *[Total Marks: 8]*

18 **(a)** Reflection **[B1]**

In the line $y = -x$ **[B1]**

(b) Correct enlargement drawn with vertices at $(0, 0)$, $(2, 0)$, $(2, 1)$, and $(1, 1)$ **[B3]**

Award [B2] for correct enlargement with scale factor $\frac{1}{2}$ from an incorrect centre

Award [B1] for any two vertices plotted correctly

(c) Correct rotation drawn with vertices at $(0, 2)$, $(0, 6)$, $(-2, 6)$, and $(-2, 4)$ **[B3]**

Award [B2] for correct orientation and size in wrong position, or 90° clockwise rotation about $(1, 1)$

Award [B1] for any two vertices plotted correctly

(d) $1 : 4$ **[B1]**

[Back to Question] *[Total Marks: 9]*

19 **(a)** Enlargement **[B1]**

Scale factor -0.5 (or $-\frac{1}{2}$) **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Rotation **[B1]**

90° clockwise (or 270° anti-clockwise) **[B1]**

Centre $(2, 2)$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 0.5 (or $\frac{1}{2}$) **[B1]**

Centre $(0, 0)$ **[B1]**

[Back to Question] *[Total Marks: 9]*

20 **(a)** Enlargement **[B1]**

Centre $(0, 0)$ or origin **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

(b) Image R drawn correctly at $(1, 3), (3, 3), (3, 4)$ **[B3]**

If incorrect, 2 marks for correct size and orientation but wrong position **[M2]**

1 mark for correct centre used but wrong scale factor **[M1]**

(c) Rotation **[B1]**

$180°$ **[B1]**

Centre $(0, 1)$ **[B1]**

OR Enlargement [B1], Scale factor -1 [B1], Centre $(0, 1)$ **[B1]**

[Back to Question] *[Total Marks: 9]*

21 **(a)** Enlargement **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Back to Question] *[Total Marks: 5]*

22 **(a)** Correct triangle U with vertices at $(1, 2), (3, 2)$ and $(3, 0)$ **[B2]**

Enlargement scale factor $\frac{1}{2}$ from wrong centre OR ray lines drawn from $(0, 4)$ but scale factor incorrect **[B1]**

(b) Correct triangle V with vertices at $(2, 1), (2, 3)$ and $(0, 3)$ **[B2]**

Reflection of U in the line $y = -x$ OR reflection of T in the line $y = x$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 2 **[B1]**

Centre $(0, 4)$ **[B1]**

[Back to Question] *[Total Marks: 7]*

23 **(a)** Correct triangle drawn with vertices at $(-1, -2)$, $(-7, -2)$, $(-7, -6)$ **[B3]**

If not B3, B2 for a correct enlargement of SF -2 but wrong centre **[B2]**

If not B2, B1 for enlargement of SF 2 from centre $(1, 0)$ OR two correct vertices **[B1]**

(b) $(-1, -2)$ **[B1]**

$(-1, -5)$ and $(-3, -5)$ **[B1]**

(c) Rotation **[B1]**

90° clockwise OR 270° anticlockwise **[B1]**

Centre $(7, 3)$ **[B1]**

[Back to Question] *[Total Marks: 8]*

24 **(a)** Subtracts $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ from at least one vertex of U **[M1]**

$(2, 4), (5, 4), (4, 8)$ **[A1]**

(b) Uses the transformation (x, y)

(y, x) $for at least one coordinate [M1]$

$(2, 5), (2, 8), (6, 7)$ **[A1]**

(c) Identifies $(2, 5)$ maps to itself **[B1]**

Applies rotation (x, y)

(y, -x) $relative to the centre (2,5) [M1]$

$(2, 5), (5, 5), (4, 1)$ **[A1]**

(d) Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 0)$ **[B1]**

[Back to Question] *[Total Marks: 10]*